



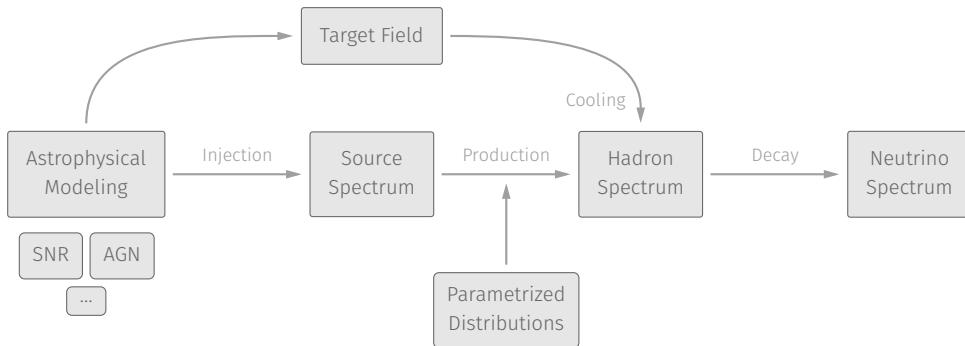
Neutrinos via Charm Decays in Astrophysical Sources

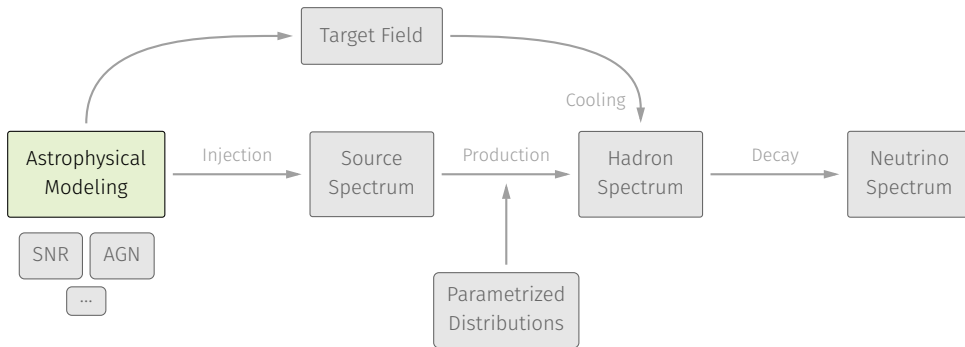
TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

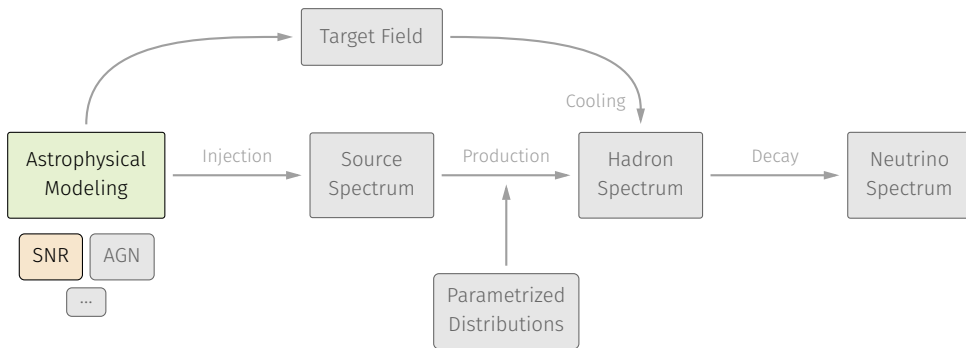
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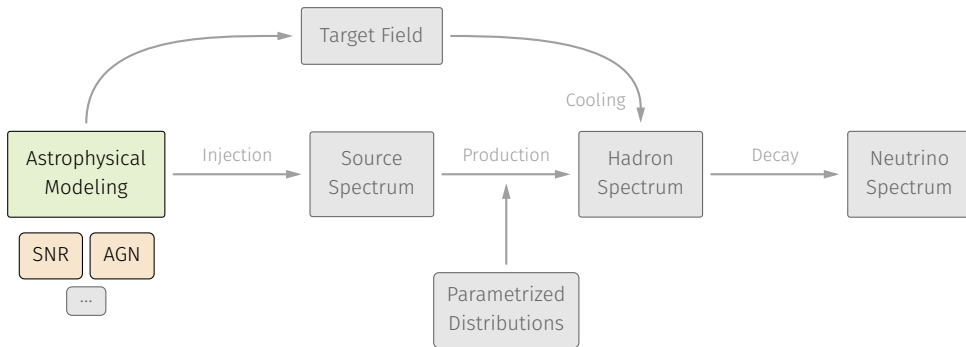
September 3, 2024

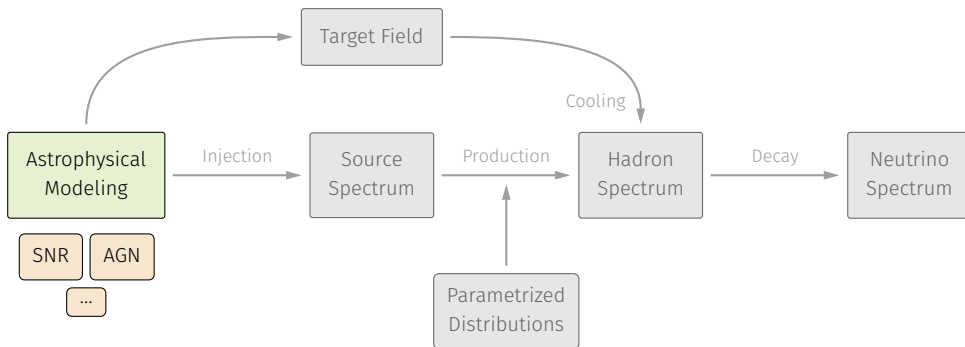
Technical University of Dortmund & Ruhr University Bochum

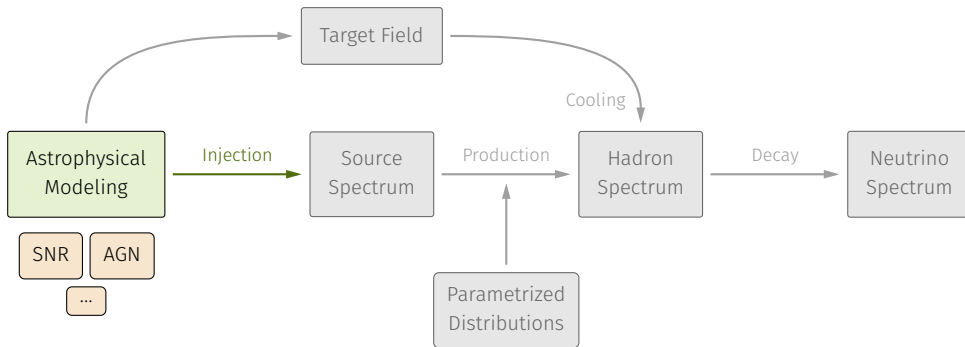


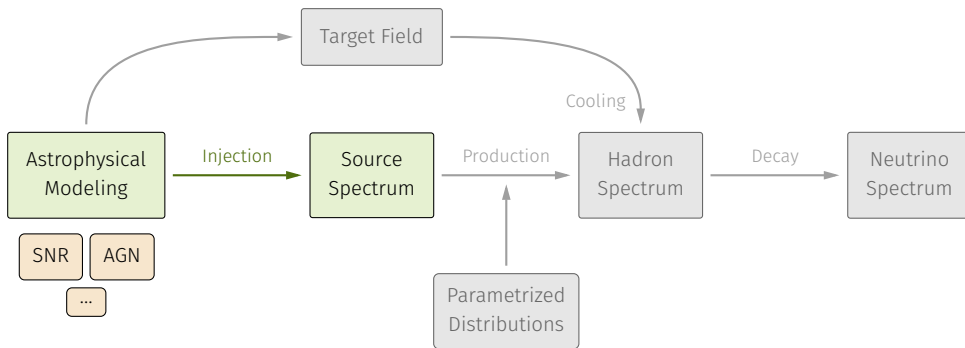


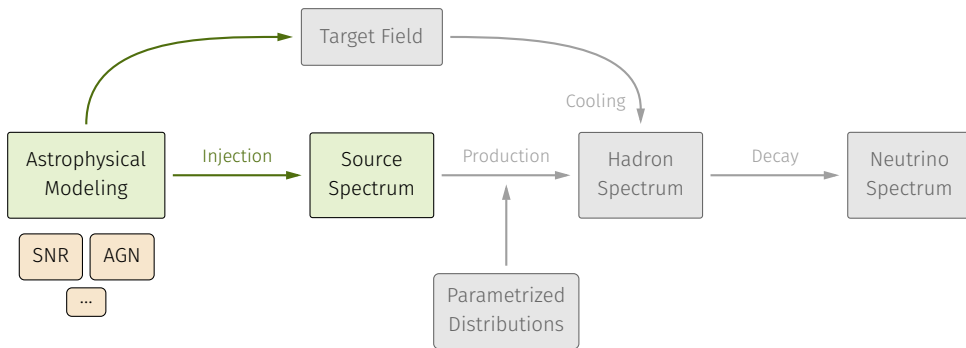


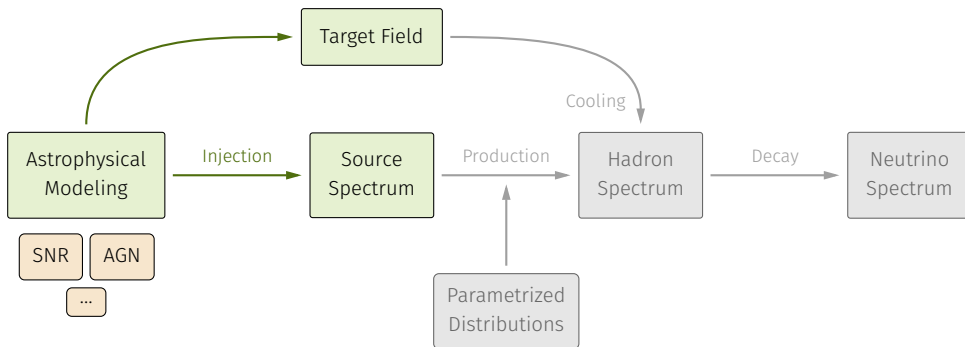


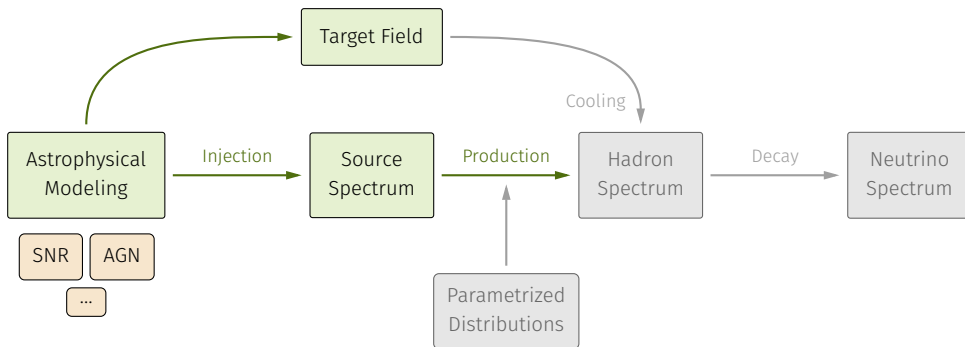


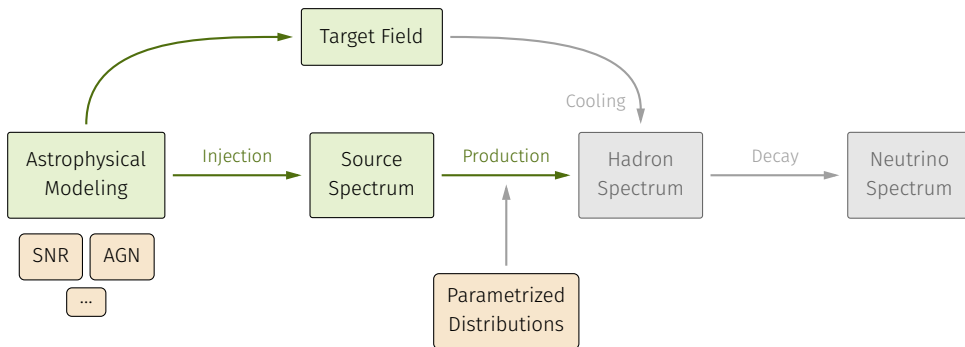


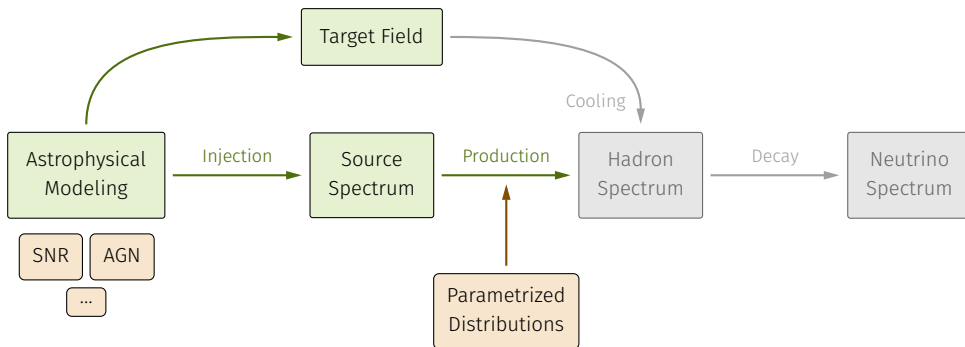


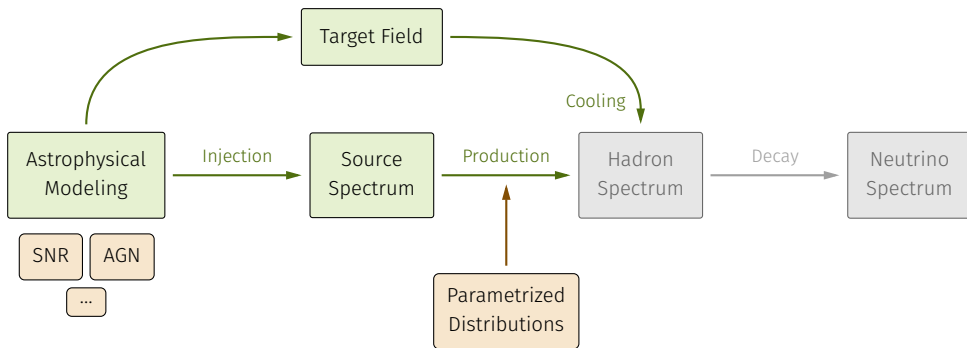


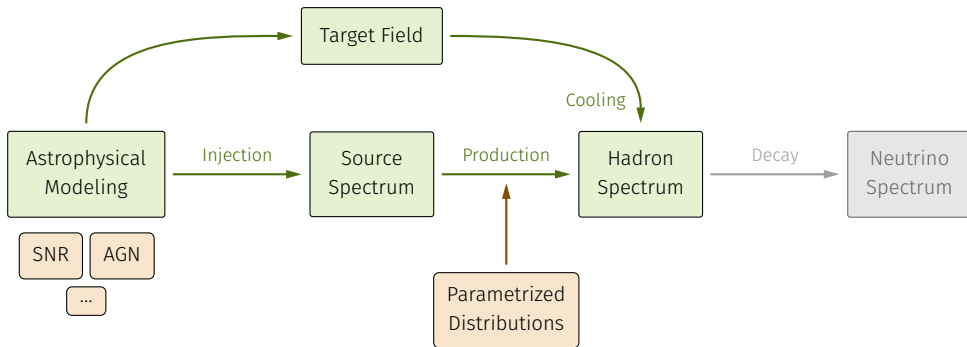


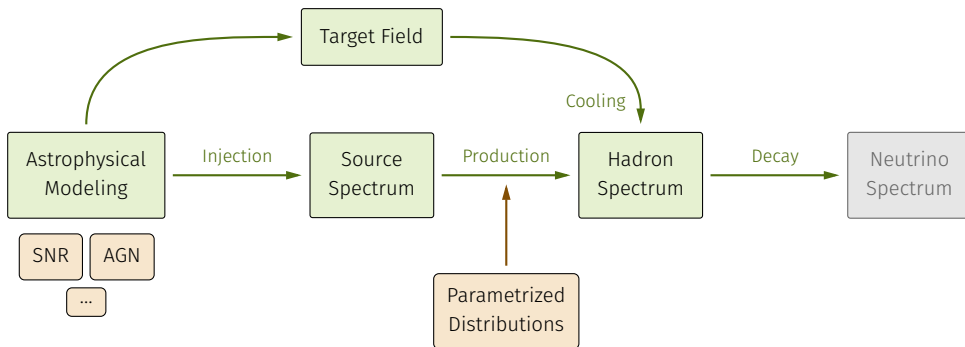


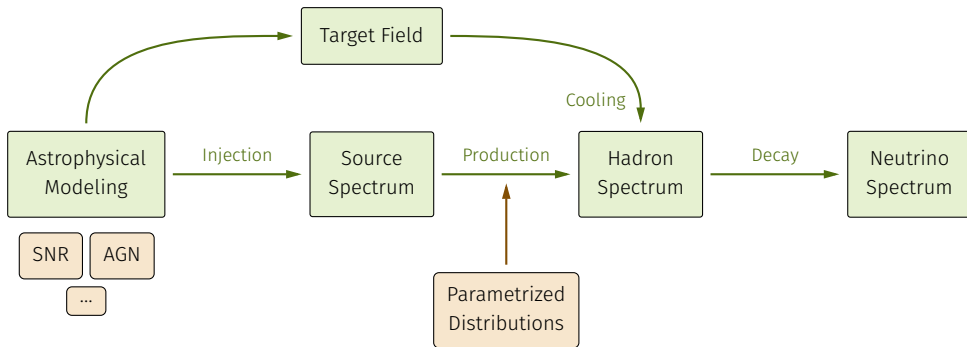


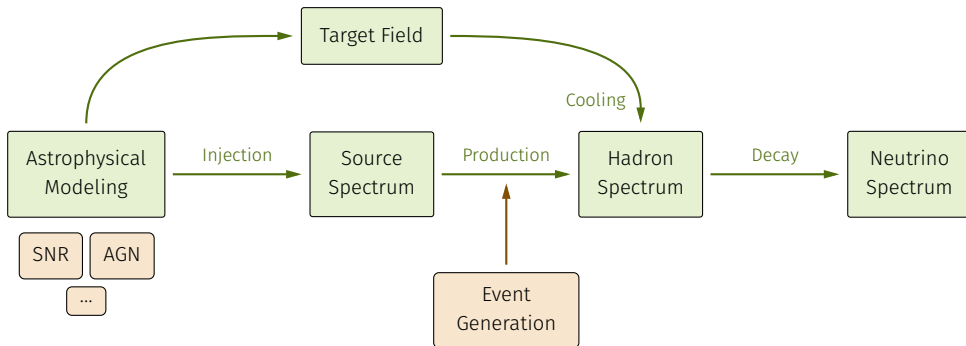




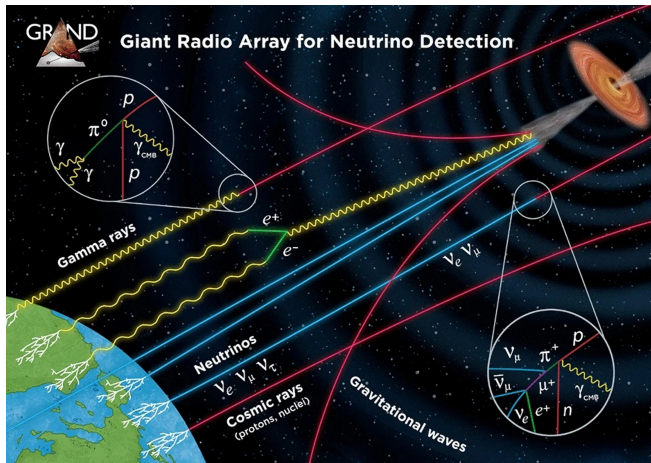






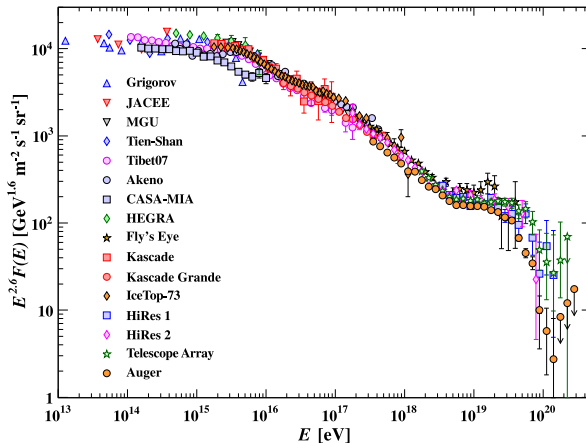


Multimessenger Astrophysics



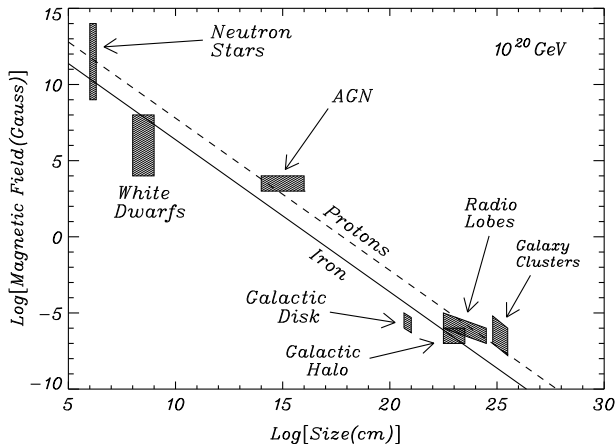
X. L. Qian et. al.
Nucl. Sci. Tech. (2021)

Cosmic Ray Spectrum & Cutoff



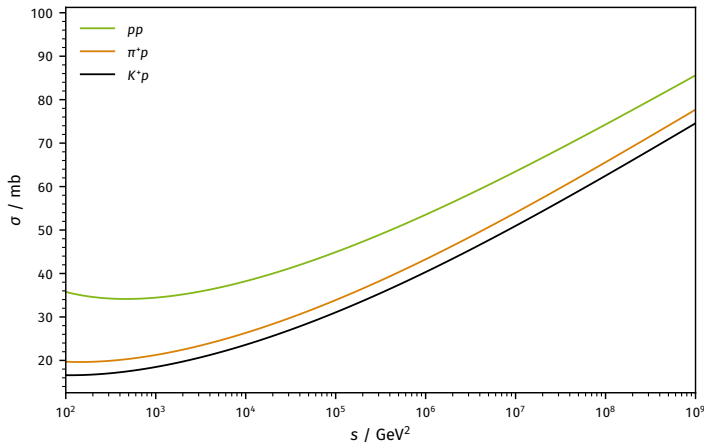
K. A. Olive et. al.
Chin. Phys. C (2014)

Possible High Energy Cosmic Ray Origins



A. V. Olinto
AIP Conf. Proc. (2000)

Hadron Inelastic Scattering Cross Section

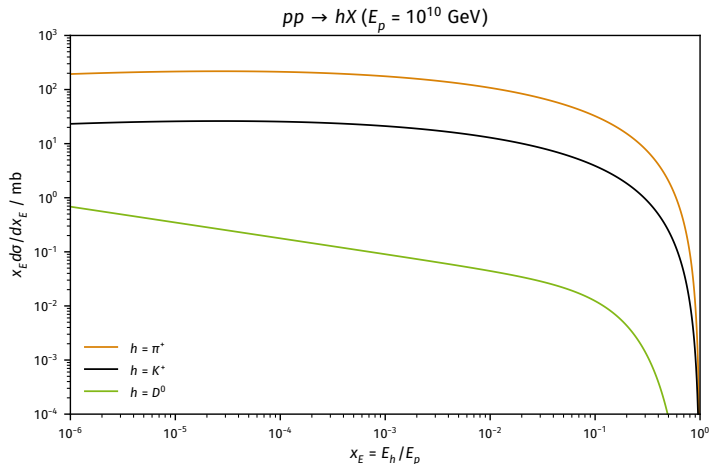


Combined Parametrizations

V. I. Belousov et. al.
Phys. Atom. Nuclei (2016)

D. A. Fagundes, M. J. Menon
Nucl. Phys. A (2012)

Weighted Energy Differential Cross Section

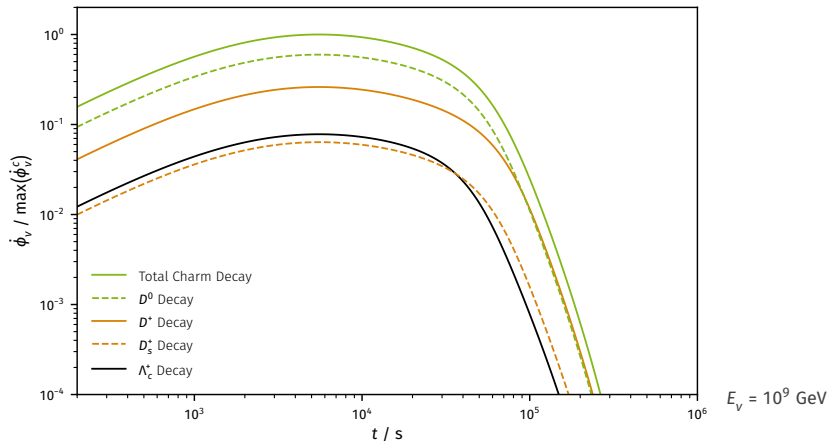


Derived Parametrizations

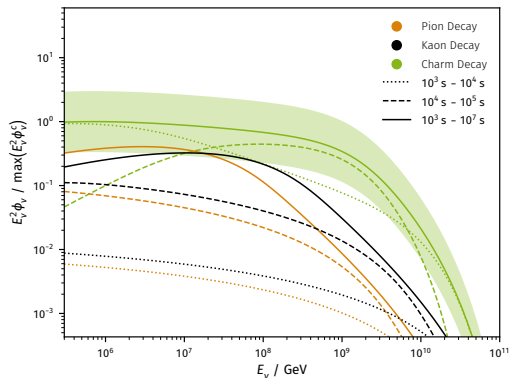
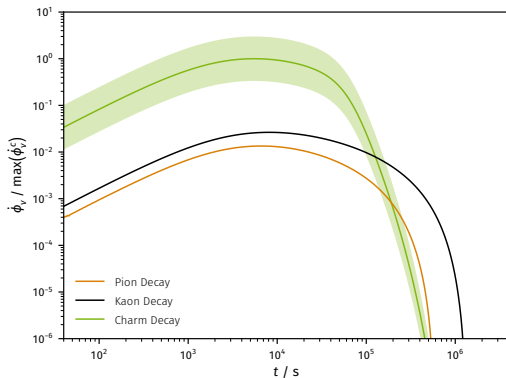
S. R. Kelner et al.
Phys. Rev. D (2006)

V. P. Gonçalves et. al.
JHEP (2007)

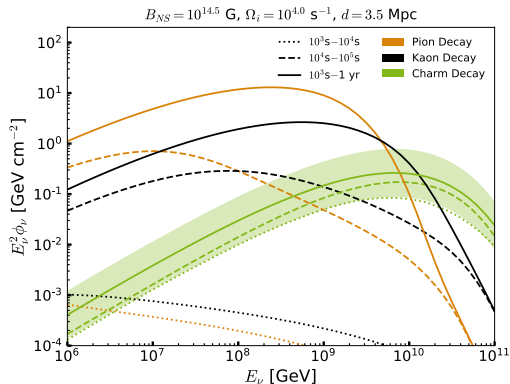
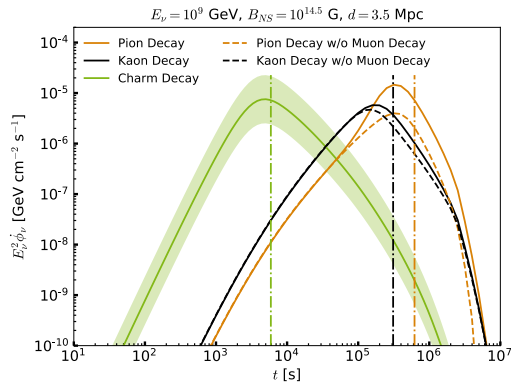
Normalized Magnetar Charm Contributions



Normalized Magnetar Neutrino Flux & Fluence

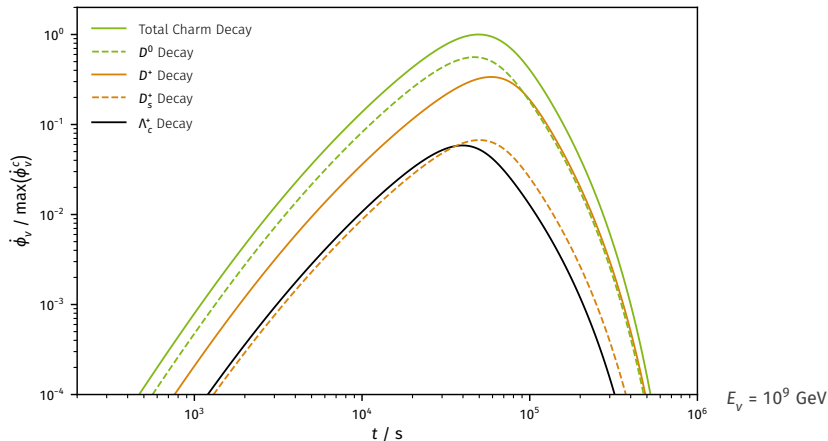


Comparison: Published Estimated Magnetar Neutrinos

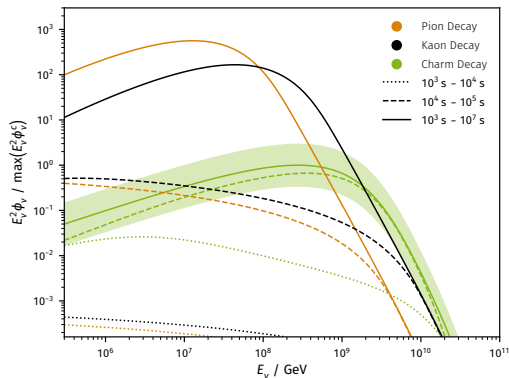
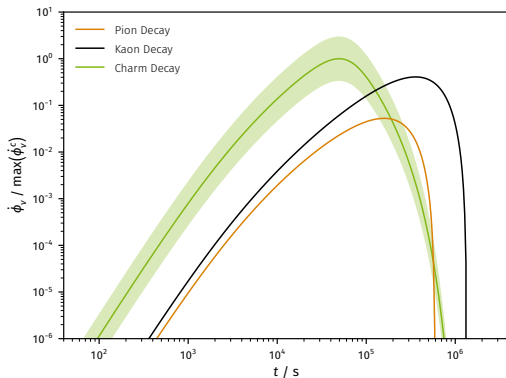


J. A. Carpio et. al. Phys. Rev. D (2020)

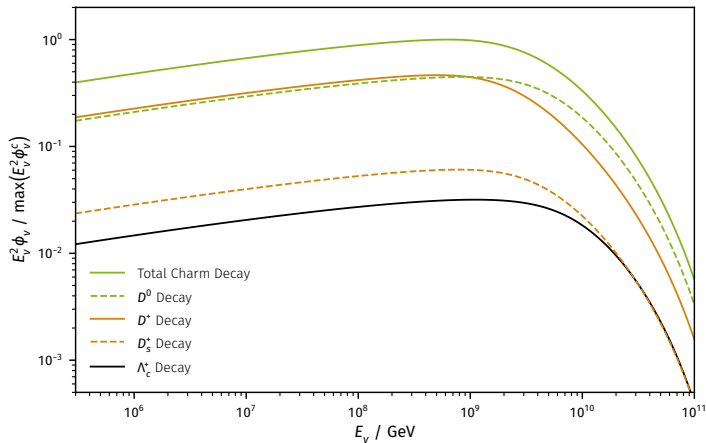
Modified Magnetar Charm Contributions



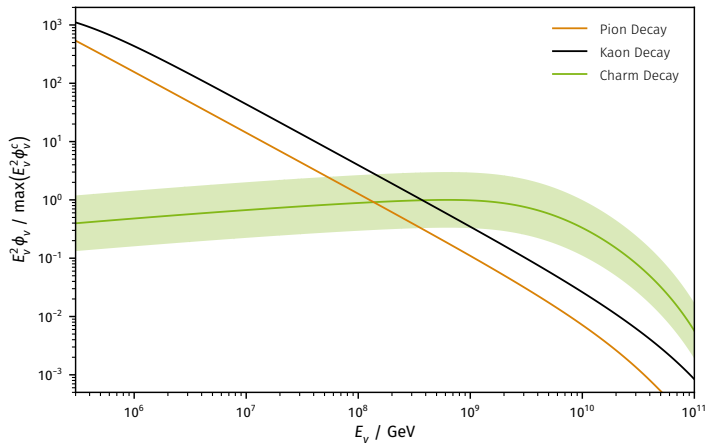
Modified Magnetar Neutrino Flux & Fluence



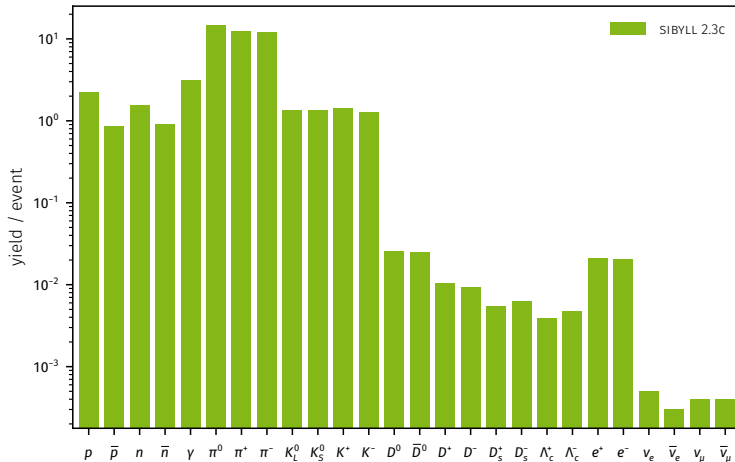
Normalized AGN Charm Contributions



Normalized AGN Neutrino Fluence

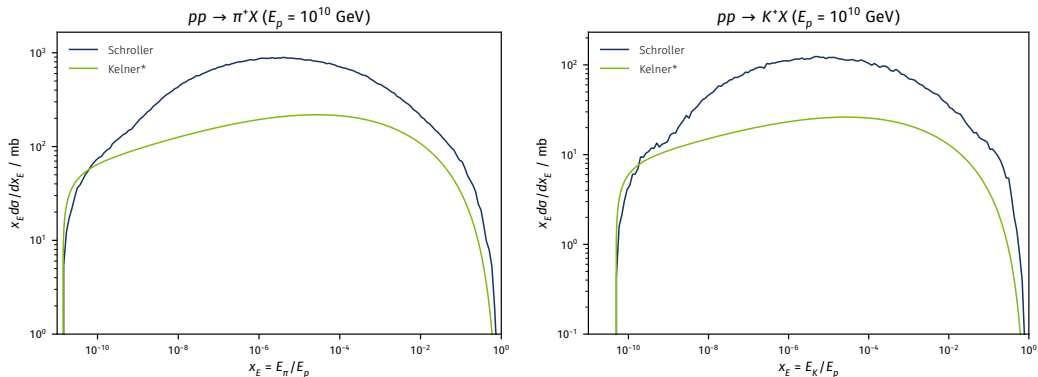


Monte Carlo Event Generation



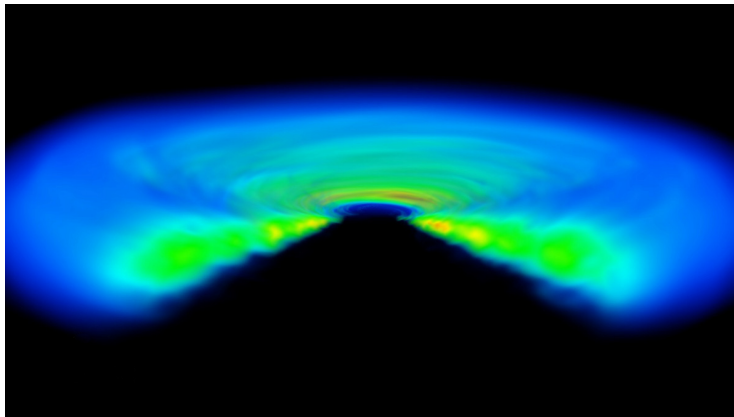
Data Courtesy
M. Schroller

Comparison: Differential Cross Section



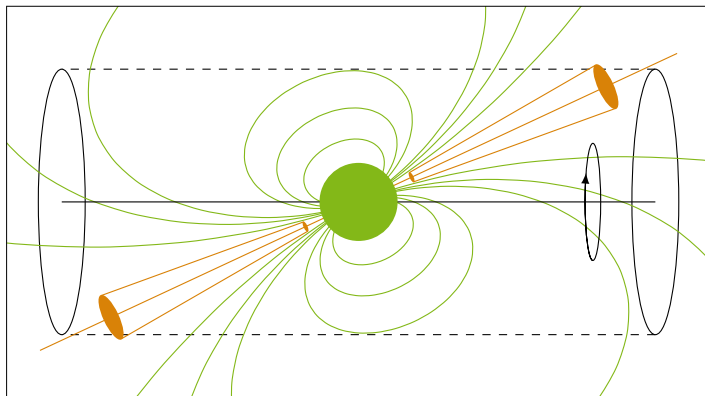
Derived Parametrization S. R. Kelner et. al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schrollier

Simulated Accretion Disk Logarithmic Density



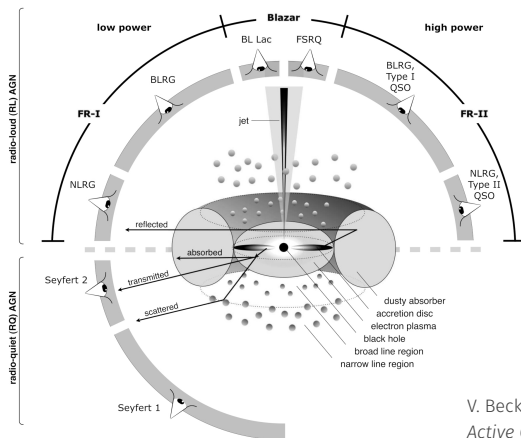
V. V. Zhuravlev et. al.
ApJ (2014)

Pulsar



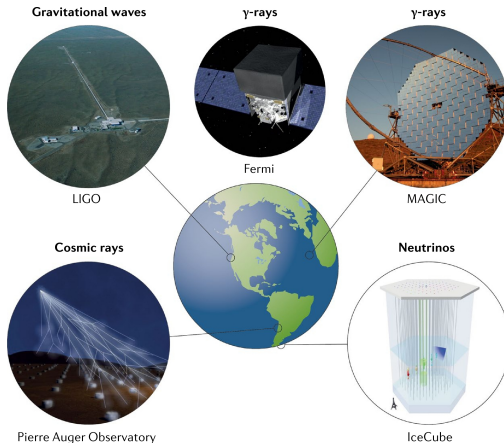
A. Knierim
Zenodo (2024)

Unified AGN Classification Model



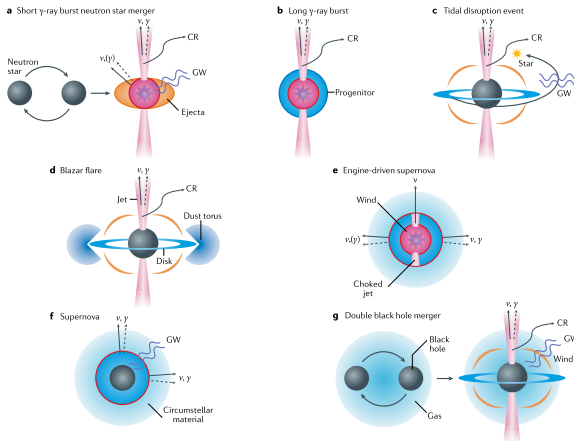
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Detectors



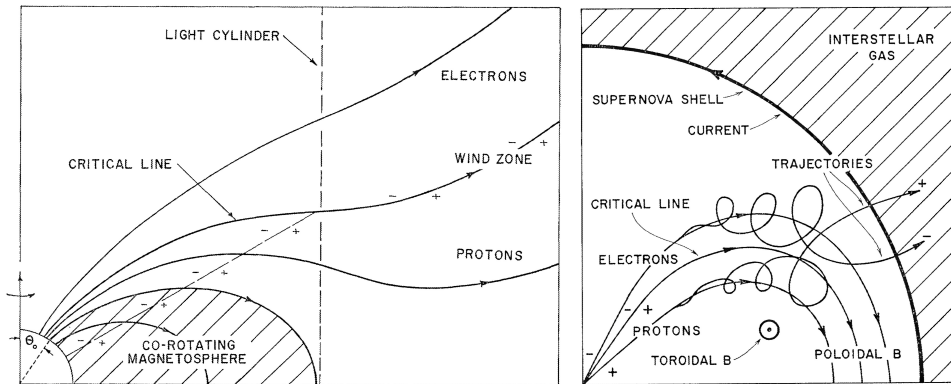
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Sources



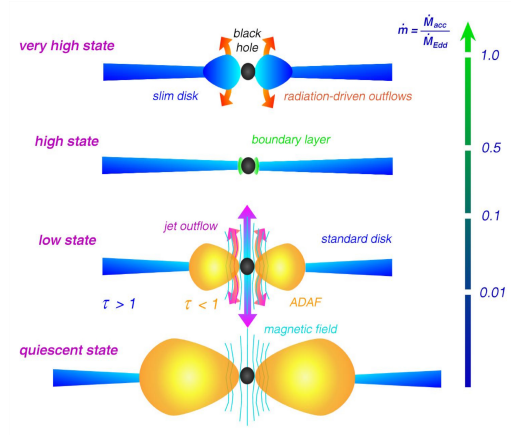
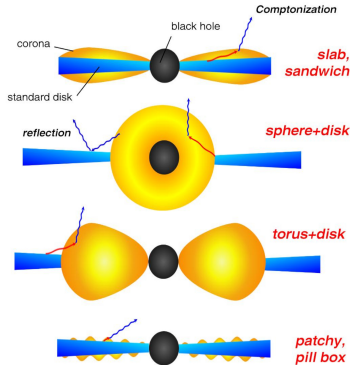
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

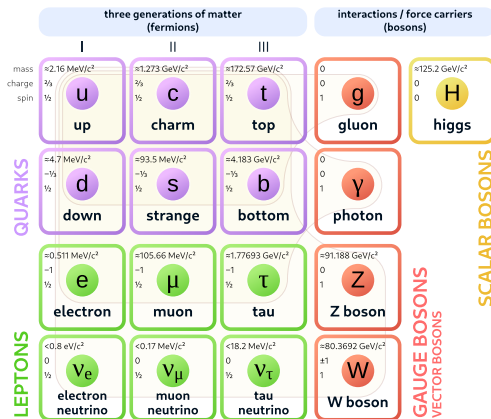
AGN Accretion Disk & Corona



A. Müller University Heidelberg (2004)

Elementary Particles

Standard Model of Elementary Particles



Standard Model
Wikimedia Commons (2019)